Reference Desk Subclass Reach

Assignment Problem 2

Concept and approach

Add two subclass contexts. For an access from a subclass in a different package, protected access works through a reference whose declared type is the subclass (OWN_TYPE), but not through a plain parent-typed reference (PARENT_TYPE). private and package-private remain denied for both contexts.

Important implementation notes

The loop returns immediately at the first denied attempt; later attempts are not checked. The context strings must match the assignment spelling exactly.

Java solution

```
class AccessChecker {
static String classifyAccess(String modifier, String context) {
switch (modifier) {
case "private":
return context.equals("SAME_CLASS") ? "ALLOWED" : "DENIED";
case "default":
return context.equals("SAME_CLASS") || context.equals("SAME_PACKAGE")
? "ALLOWED" : "DENIED";
case "protected":
if (context.equals("SAME_CLASS") || context.equals("SAME_PACKAGE") ||
context.equals("SUBCLASS_DIFFERENT_PACKAGE_OWN_TYPE")) return "ALLOWED";
return "DENIED";
case "public":
return "ALLOWED";
default:
throw new IllegalArgumentException("Unknown modifier: " + modifier);
}
}

static String firstDeniedAttempt(String[][] attempts) {
for (int i = 0; i < attempts.length; i++) {
String modifier = attempts[i][0];
String context = attempts[i][1];
if (classifyAccess(modifier, context).equals("DENIED"))
return modifier + " via " + context + " (attempt #" + (i + 1) + ")";
}
return "None Denied";
}
}
```

Sample output

```
protected via SUBCLASS_DIFFERENT_PACKAGE_PARENT_TYPE (attempt #2)
```